

Metrics Verification Issues

Missing Metrics

1. Win Streak Metrics Discrepancy:

- `metrics_calculator.py` calculates win_streak_2, win_streak_3, win_streak_4, win_streak_5
- `bot_ranker.py` expects win_streak_2, win_streak_3, win_streak_4, win_streak_5, win_streak_6, win_streak_7
- Need to add calculation for win_streak_6 and win_streak_7 in `metrics_calculator.py`

2. Model Score Metrics Missing:

- `bot_ranker.py` expects: price_model_score, volume_model_score, price_wall_score
- These don't appear to be calculated anywhere in `metrics_calculator.py`
- Need to implement calculation methods for these metrics or adjust `bot_ranker.py` to handle their absence

Data Storage Issues

1. Algorithm ID Hardcoded:

- In `metrics_updater.py`, algo_id is hardcoded to 1 with a comment to replace with actual logic
- Need to implement proper algo_id determination logic

2. Database Constraint Issues:

- `metrics_updater.py` uses "ON CONFLICT (bot_id, algorithm_id)" suggesting both together form a unique key
- `bot_ranker.py` uses "ON CONFLICT (bot_id)" when updating rankings, suggesting bot_id alone is unique
- Need to verify database schema constraints match the assumptions in the code

3. Update Implementation for Win Streaks:

- Win streaks are updated in a separate function call in `metrics_updater.py`
- This could lead to race conditions or inconsistent data if there's an error between updates
- Consider consolidating all updates into a single SQL statement

Query Logic Issues

1. Inconsistent Time Periods:

- Some performance metrics use NOW() in SQL queries, others pass datetime objects
- Standardize the approach to time period calculations

2. Default Values:

- Some functions use `result` or `0.0` for NULL handling, others don't have explicit NULL handling
- Implement consistent NULL handling across all metric calculations

General Problems

1. Error Handling:

- Most functions don't have specific error handling for database operations
- Add more specific error handling and logging

2. Metrics Validation:

- No validation that metric values are within reasonable ranges
- Add validation to ensure calculated metrics make sense

To-Do List

1. Update `metrics_calculator.py` to calculate all metrics expected by `bot_ranker.py`
2. Update `metrics_updater.py` to store all required metrics
3. Implement proper `algo_id` determination
4. Review and update database schema to ensure it matches code assumptions
5. Improve error handling and add validation
6. Add automated tests for metric calculations